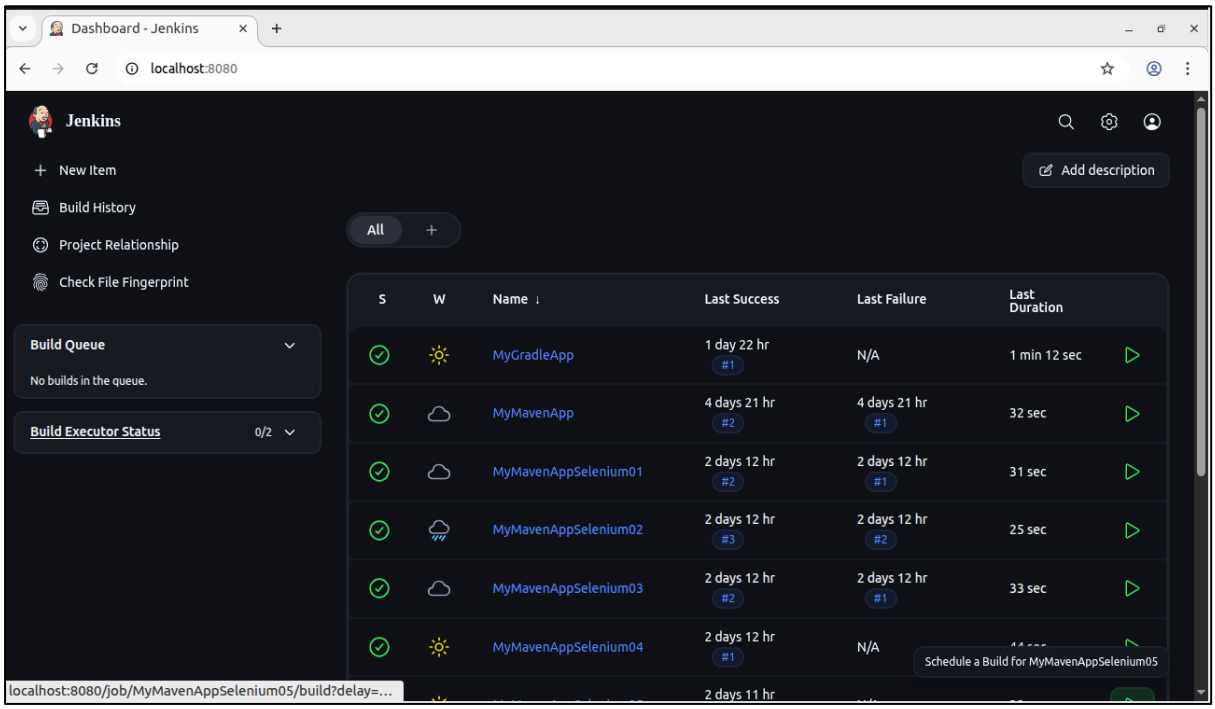
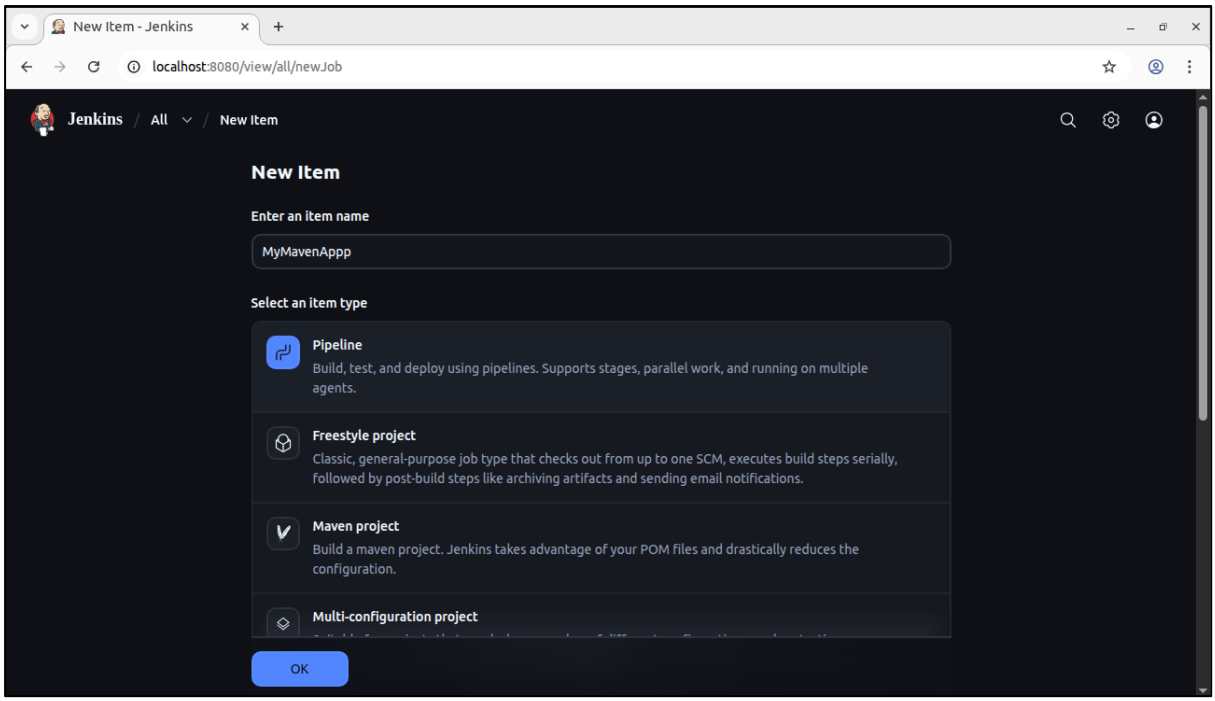
Jenkins Pipeline for MyMavenApp

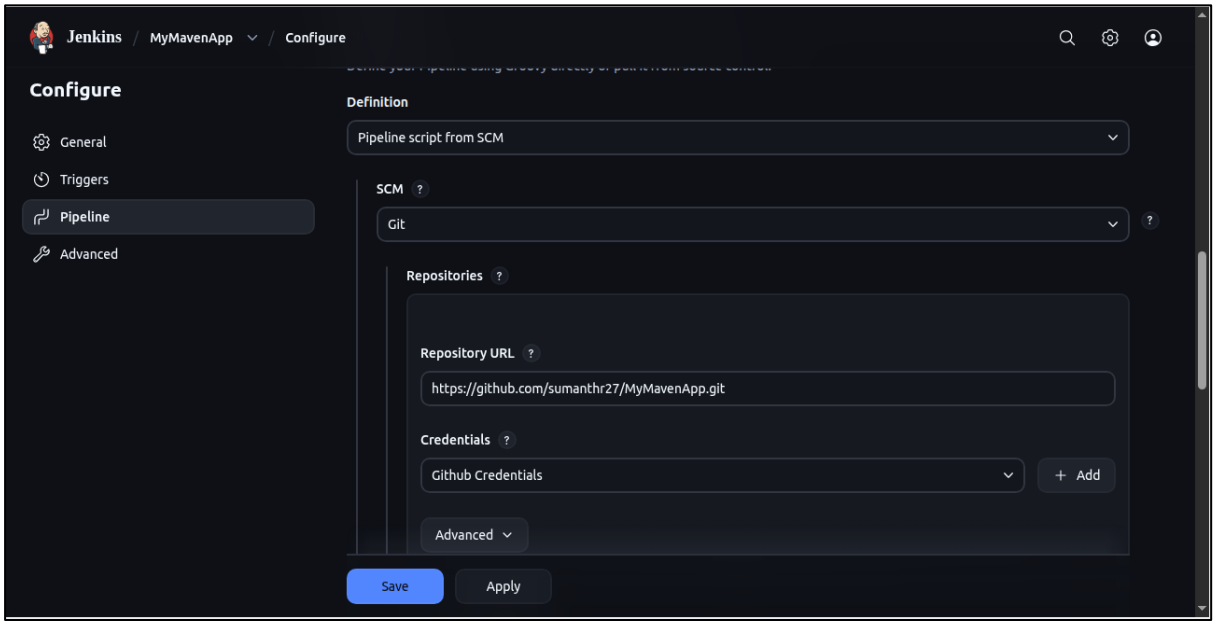
Step 1: Open the Jenkins Dashboard in order to create a new item



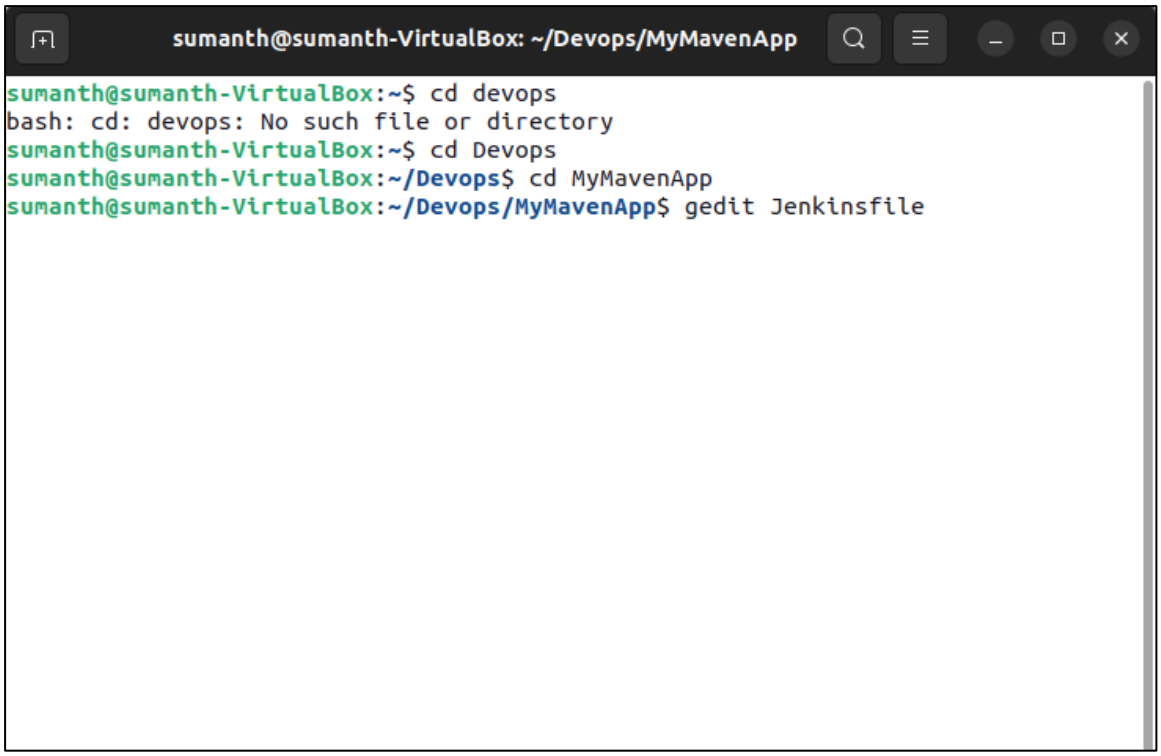
Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenApp” and Item type as Pipeline and Create it



Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenApp and create an new file called “Jenkinsfile”



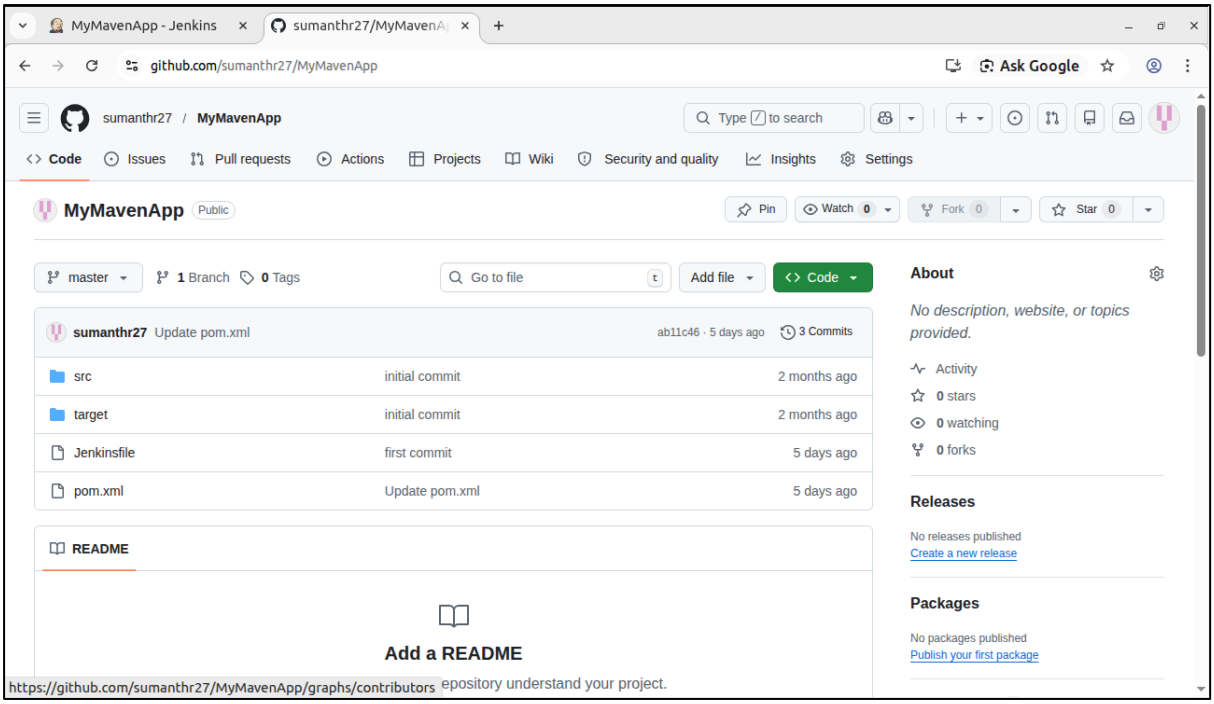
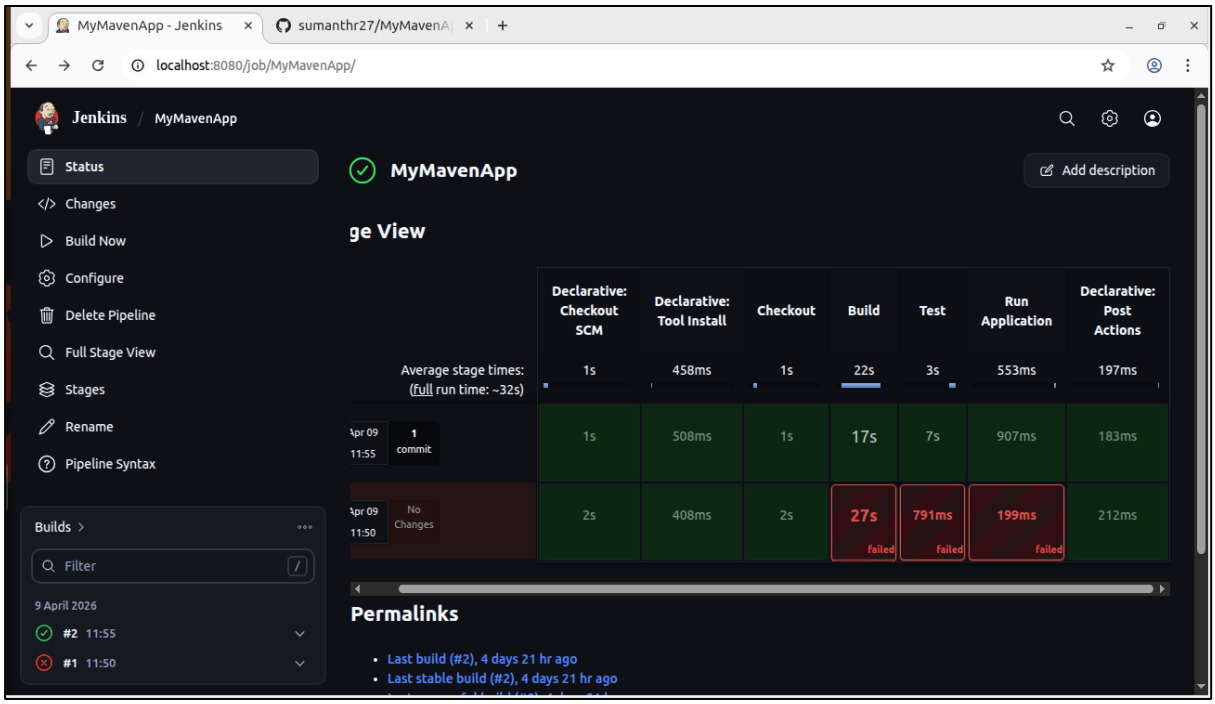
Step 5: Paste the available code inside the Jenkinsfile

```
pipeline {
    agent any  // Use any available agent
    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/sumanthr27/MyMavenApp.git'
            }
        }
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }
        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
            }
        }
    }
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit ny pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.